Hints: Integrals – 1. Substitutions: rationalisation, Euler, the half-angle substitution

Class work

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. For each Euler substitution, square the defining relation: e.g. $ax^2+bx+c=(\sqrt{a}\,x+t)^2$ makes the x^2 terms cancel, so x is found from a *linear* equation in x and is rational in t; similarly $y=xt\pm\sqrt{c}$ kills the constant term (divide by x), and $y=t(x-x_1)$ lets you cancel the factor $x-x_1$. The first needs $a>0$, the second $c>0$, the third real roots; pick the one whose resulting rational function has the lowest degree.

Hint 2. For $\int \frac{dx}{x\sqrt{x^2+4x+3}}$ you can use $y=xt+\sqrt{3}$ (since $c=3>0$), or $y=t(x+1)$ (roots $-1,-3$). A faster route: $x=1/u$ turns it (for $x>0$) into $-\int \frac{du}{\sqrt{3u^2+4u+1}}$, a standard log integral. For $x>0$ the answer is $-\frac{1}{\sqrt{3}}\ln\left|\frac{2x+3+\sqrt{3}\sqrt{x^2+4x+3}}{x}\right|+C$.

Hint 3. For $\int\sqrt{x^2+1}\,dx$: Euler $\sqrt{x^2+1}=t-x$ gives $x=\frac{t^2-1}{2t}$, $\sqrt{x^2+1}=\frac{t^2+1}{2t}$; hyperbolic $x=\sinh u$ gives $\int\cosh^2 u\,du$. Both lead to $\frac{1}{2}\left(x\sqrt{x^2+1}+\ln(x+\sqrt{x^2+1})\right)+C$; note $t=x+\sqrt{x^2+1}=e^u$, which explains why the answers agree.

Problem 2

Hint 1. Write $\sin x=2\sin\frac{x}{2}\cos\frac{x}{2}$, $\cos x=\cos^2\frac{x}{2}-\sin^2\frac{x}{2}$, divide by $1=\cos^2\frac{x}{2}+\sin^2\frac{x}{2}$, and divide numerator and denominator by $\cos^2\frac{x}{2}$. For dx differentiate $x=2\arctan t$.

Hint 2. $\int\frac{dx}{2+\cos x}=\int\frac{2\,dt}{t^2+3}=\frac{2}{\sqrt{3}}\arctan\frac{\tan(x/2)}{\sqrt{3}}+C$ (on each interval of continuity of $\tan\frac{x}{2}$). $\int\frac{dx}{1+\sin x}=\int\frac{2\,dt}{(1+t)^2}=-\frac{2}{1+\tan(x/2)}+C$, which equals $\tan x-\sec x$ up to a constant (or multiply by $1-\sin x$ directly).

Hint 3. For the third: $\frac{1}{\sin x+\tan x}=\frac{\cos x}{\sin x(1+\cos x)}$ is odd in $\sin x$, so $t=\cos x$ works after multiplying top and bottom by $\sin x$: you get $-\int\frac{t\,dt}{(1-t)(1+t)^2}$ and partial fractions. Now compare with $t=\tan\frac{x}{2}$: $\sin x+\tan x=\frac{4t}{(1+t^2)(1-t^2)}$, and the factor $1+t^2$ cancels against $dx=\frac{2\,dt}{1+t^2}$, so you get the much simpler $\int\frac{1-t^2}{2t}\,dt=\frac{1}{2}\ln|\tan\frac{x}{2}|-\frac{1}{4}\tan^2\frac{x}{2}+C$, with no partial fractions at all. That is why the half-angle substitution wins here.

Problem 3

Hint 1.

(a) needs no substitution: expand the product. With $a=x^{1/2}$, $b=x^{1/3}$ it equals $3+a+\frac{1}{a}+b+\frac{1}{b}+\frac{a}{b}+\frac{b}{a}$, and $\frac{a}{b}=x^{1/6}$; integrate each power. (c) simplifies first: $x+\sqrt{x}=\sqrt{x}(\sqrt{x}+1)$.

Hint 2. (b), (d): put $x=t^2$, $dx=2t\,dt$. (b) becomes $\int\frac{2t\,dt}{(t+2)^2+1}$, giving $\ln(x+4\sqrt{x}+5)-4\arctan(\sqrt{x}+2)+C$. (d) becomes $\int\frac{2\,dt}{t+3}=2\ln(\sqrt{x}+3)+C$. (c) is $\frac{2}{3}x^{3/2}+C$.

Problem 4

Hint 1. Look for the derivative of an inner function: (a) $u=x^3+2$; (b) $u=x^2+1$, writing $x^3=x^2\cdot x$; (c) $u=x^2$ (then complete the square in u^2+u+1); (d) multiply by x/x and put $u=x^2$, or use $\frac{1}{x(x^2+1)}=\frac{1}{x}-\frac{x}{x^2+1}$; (e) $u=x^6$.

Hint 2. Answers: (a) $\frac{2}{9}(x^3+2)^{3/2}$; (b) $\frac{1}{5}(x^2+1)^{5/2} - \frac{1}{3}(x^2+1)^{3/2}$; (c) $\frac{1}{\sqrt{3}}\arctan\frac{2x^2+1}{\sqrt{3}}$; (d) $\ln|x| - \frac{1}{2}\ln(x^2+1)$; (e) $\frac{1}{6\sqrt{2}}\arctan\frac{x^6}{\sqrt{2}}$ (all $+C$).

Hint 3. (f): put $x = t^4$. Then $x\sqrt{x} - x^2 = t^6 - t^8 = t^6(1-t^2)$ and $dx = 4t^3 dt$, so the integral is $\int \frac{4\,dt}{\sqrt{1-t^2}} = 4\arcsin(x^{1/4}) + C$.

Problem 5

Hint 1. Define the infinite radical as the limit of the recursion $y_{n+1} = \sqrt{x + y_n}$ (resp. $y_{n+1} = \sqrt{x\,y_n}$), $y_0 = 0$ or $y_0 = \sqrt{x}$. Show the sequence is monotone and bounded, so the limit y exists and satisfies the fixed-point equation $y = \sqrt{x+y}$ (resp. $y = \sqrt{xy}$), and take the positive root.

Hint 2. First: $y^2 - y - x = 0$, $y = \frac{1+\sqrt{1+4x}}{2}$, so the integral is $\frac{x}{2} + \frac{1}{12}(1+4x)^{3/2} + C$. Second: $y = x^{1/2+1/4+\cdots} = x$ for $x > 0$, so the answer is $\frac{x^2}{2} + C$.

Hint 3. Third: $y^2 - y - (x^2+x) = 0$ gives $y = \frac{1+\sqrt{(2x+1)^2}}{2} = x+1$ for $x \geqslant 0$. Hence $\int_0^1 (x+1)\,dx = \frac{3}{2}$.